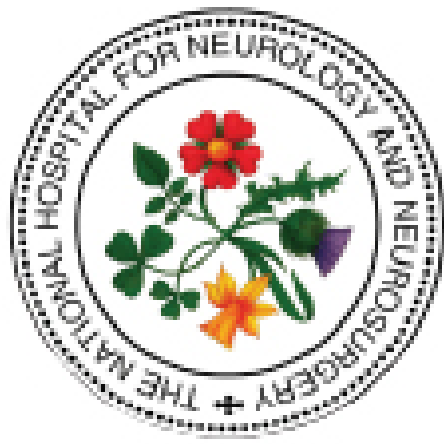




MILRINONE for VASOSPASM in treated Aneurysmal Subarachnoid Haemorrhage

SUSPECTED DELAYED CEREBRAL ISCHAEMIA (DCI)

SIGNS AND SYMPTOMS

Sedated Patient
Raised ICP
Self-hypertensing
Vasospasm on TCD

Awake Patient
Fluctuating GCS
Headache
Self-hypertensing
Focal neurological deficit
Agitation
Vasospasm on TCD

CT SCAN TO RULE OUT OTHER CAUSES

When prescribing, select 'Milrinone infusion for cerebral vasospasm' option

STEP 1

Dilute **10mg/10ml** milrinone preparation to **50ml** with sodium chloride 0.9% or glucose 5% to a final concentration of **10mg in 50ml = 200mcg/mL**

STEP 2

Milrinone must be administered VIA CENTRAL LINE

Commence infusion at **0.5mcg/kg/min**

STEP 3

Increase by **0.1mcg/kg/min** at **2-hour intervals**

Up to maximum of **1mcg/kg/min** according to haemodynamic and clinical response – in some cases of severe refractory vasospasm maximum dose of **1.25mcg/kg/min** after MDT decision

Maximum daily dose should not exceed 1.8mg/kg

Continue at maximum tolerated infusion rate for **72h** then CTA / review

TARGETS

PbtO₂ ≥20mmHg

SBP >160mmHg

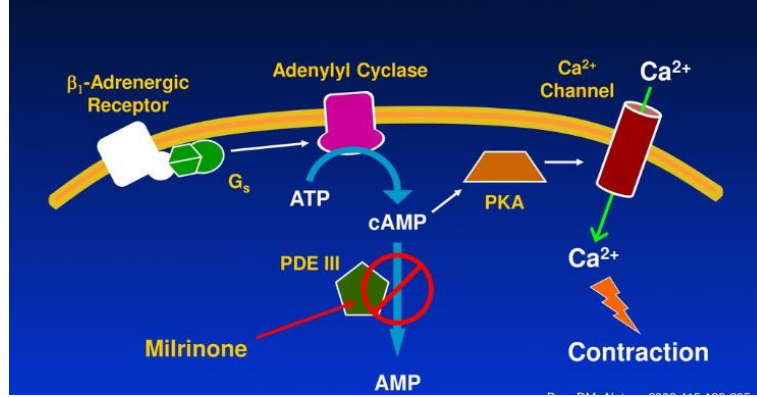
Follow PbtO₂ Protocol

in 'awake' patient **neurology** is marker of response

72 HOURS POST COMMENCEMENT OF MILRINONE

1. Repeat CTA at 72h
2. If evidence of improved vasospasm, wean milrinone by **0.25mcg/kg/min** every **24 – 48h**
3. If no improvement in vasospasm increase milrinone again as above to maximum of **1mcg/kg/min**
4. Discuss options with Consultant INR
5. Continue up to 7 days with daily review / daily TCDs

Milrinone Mechanism of Action



Selectively inhibits cAMP-specific phosphodiesterase III isoenzyme in cardiac and vascular muscle. Vasodilation occurs due to increase in cAMP in vascular smooth muscle which facilitates calcium uptake, reducing amount of calcium available for contractility. This vasodilation can directly contribute to reducing vasospasm.

PHARMACOKINETICS

Milrinone is a selective phosphodiesterase-3 inhibitor associated with circulatory inotropic support and selective cerebral-vasodilatory effects and should be considered in cerebral vasospasm with delayed cerebral ischaemia (DCI) which is refractory to standard therapy.

- Onset of action 5 -15min
- Metabolised by liver
- Half-life 2.5h
- Excreted in urine

INDICATIONS FOR USE

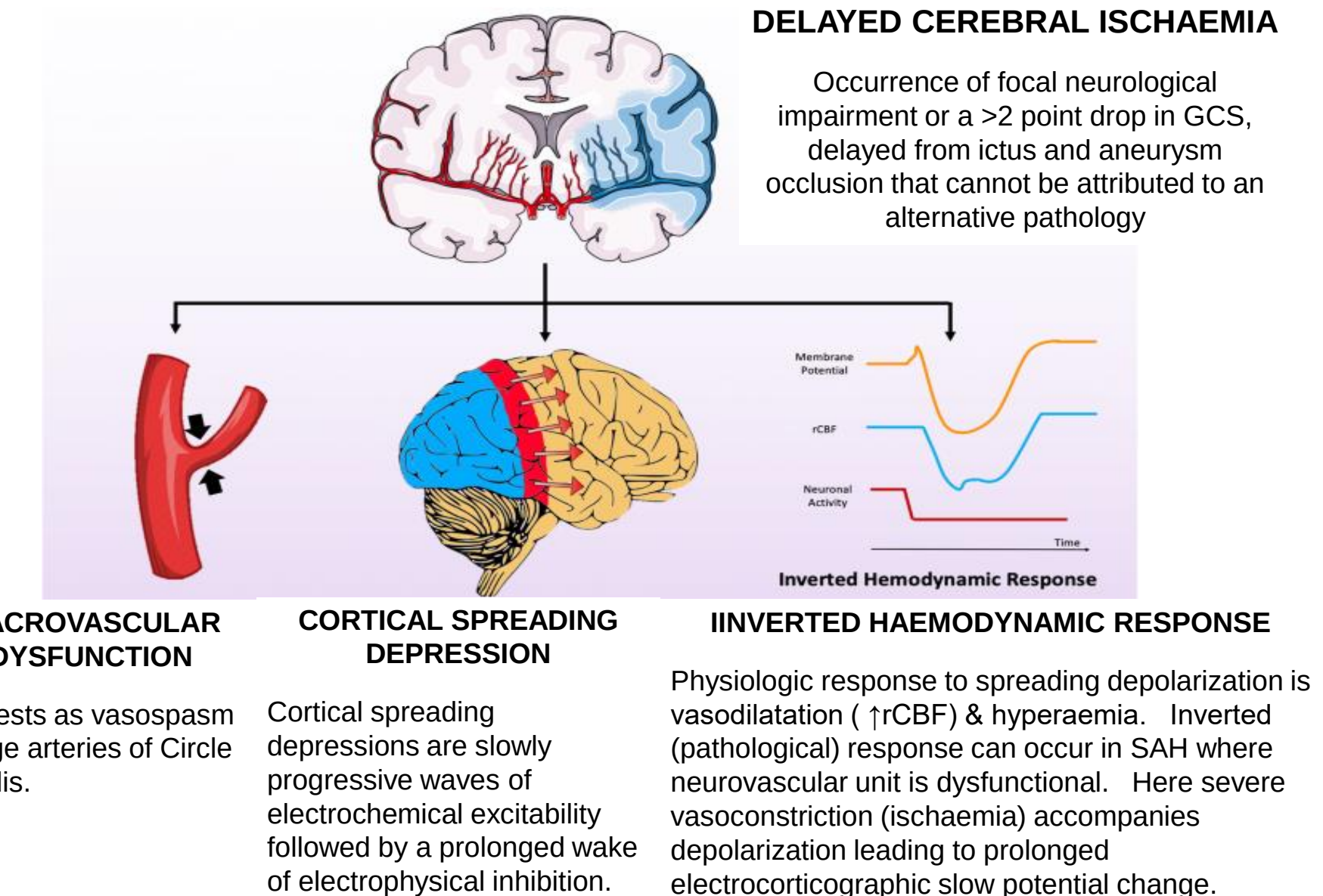
- Suspected DCI
- Refractory vasospasm confirmed with CTA, TCD or low brain oxygen tension (PbtO₂)
- Vasospasm confirmed at DSA
- Diffuse vasospasm confirmed on CTA without an interventional radiology (IR) target
- Treatment of vasospasm when IR not available
- Vasospasm with focal neurology not responsive to hypertension / angioplasty

REQUIREMENTS PRIOR TO COMMENCING MILRINONE

- Cardiac output monitoring / Central line
- Brain tissue oxygen (PbtO₂) probe in sedated patient / Clinical signs in awake patient – some awake patients may be electively sedated and intubated to facilitate PbtO₂ bolt + milrinone – *particularly if evidence of poor cardiac function and on maximum pressors*
- Noradrenaline / vasopressin infusion
- Euvolaemia
- Hb > 90g/dl
- PaO₂ > 13kPa

References

- 1.Yasser B. Abulhasan et al Milrinone for refractory cerebral vasospasm with delayed cerebral ischemia, J Neurosurg March 27, 2020
- 2.Marcelo Lannes et al Milrinone and Homeostasis to Treat Cerebral Vasospasm Associated with SAH: Neurocrit Care (2012) 16:354–362



MACROVASCULAR DYSFUNCTION
Manifests as vasospasm of large arteries of Circle of Willis.

CORTICAL SPREADING DEPRESSION
Cortical spreading depressions are slowly progressive waves of electrochemical excitability followed by a prolonged wake of electrophysiological inhibition.

INVERTED HAEMODYNAMIC RESPONSE
Physiologic response to spreading depolarization is vasodilatation (↑rCBF) & hyperaemia. Inverted (pathological) response can occur in SAH where neurovascular unit is dysfunctional. Here severe vasoconstriction (ischaemia) accompanies depolarization leading to prolonged electrocorticographic slow potential change.

IF HEART RATE increases >20bpm - reduce infusion rate to **0.25mcg/kg/min** and increase as tolerated

IN RENAL IMPAIRMENT: Dose reduction required based on creatinine clearance (CrCl) - *half-life increased in renal impairment*

ESTIMATED WEIGHT	INFUSION COMMENCING DOSE 0.5mcg/kg/min (10mg in 50ml = 200mcg/ml)
50kg	25mcg/min (7.5ml/h)
60kg	30mcg/min (9ml/h)
70kg	35mcg/min (10.5ml/h)
80kg	40mcg/min (12ml/h)
90kg	45mcg/min (13.5ml/h)
100kg	50mcg/min (15ml/h)

RENAL IMPAIRMENT : infusion rate based on creatinine clearance (CrCl)	
CrCl (ml/min/1.73m ²)	Infusion Rate (ml/kg/h)
5	0.06
10	0.07
20	0.08
30	0.10
40	0.11
50	0.13